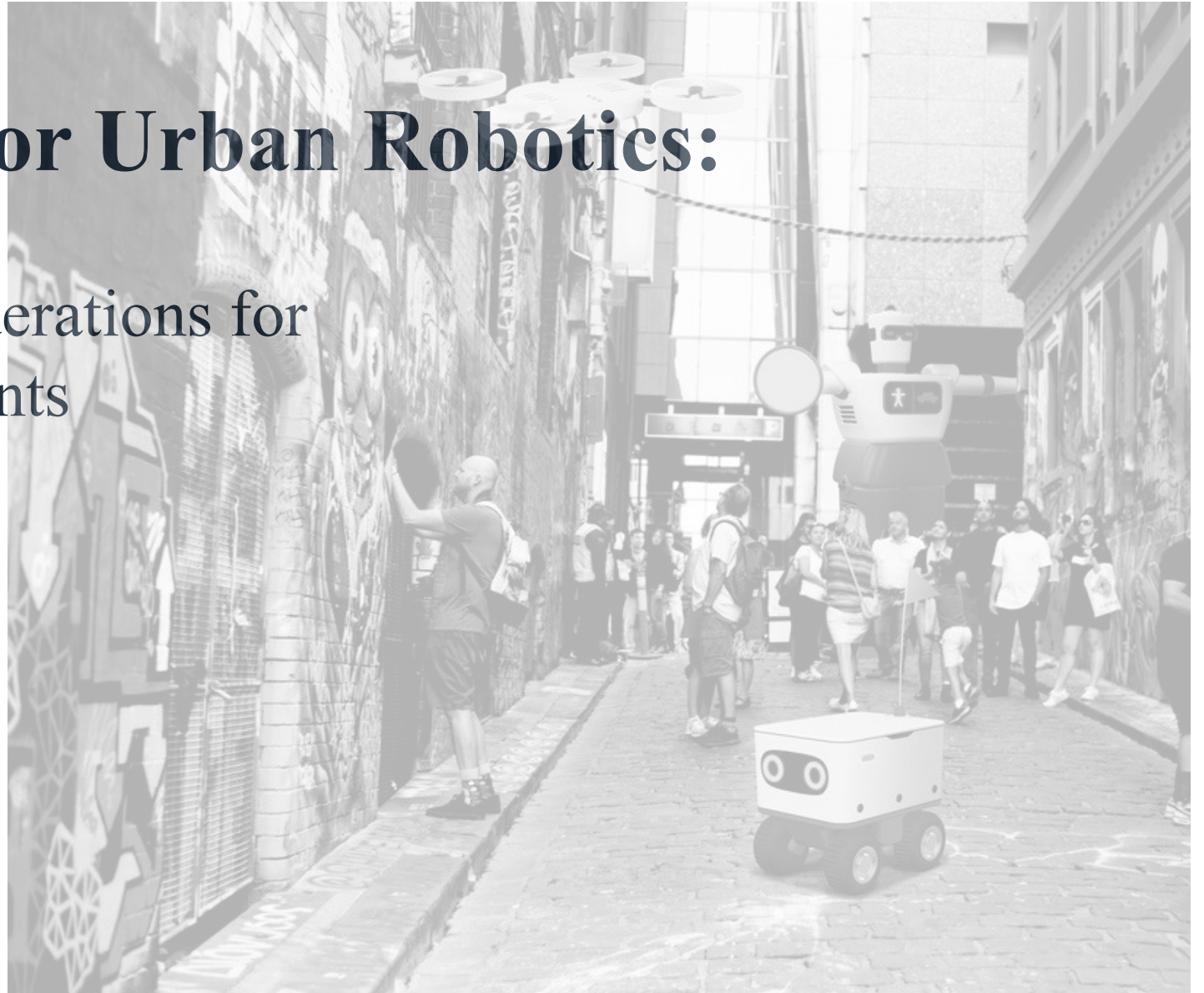


Designing Cities for Urban Robotics:

Multidimensional Considerations for
Future Urban Environments

Sung Jin Park
CEO, SPARKCITY



About Me



Sung Jin Park Ph.D.

- CEO, SPARKCITY
- Author of "Seoul: Tale of A Smart City"
- 20+ Years of Experience in Smart Cities
- Smart City Scholar and Practitioner
- Former Smart City Consultant of Samsung and LG
- Led Domestic and Global Smart City Projects
- Robot Friendly Building R&D Consultant ('25)

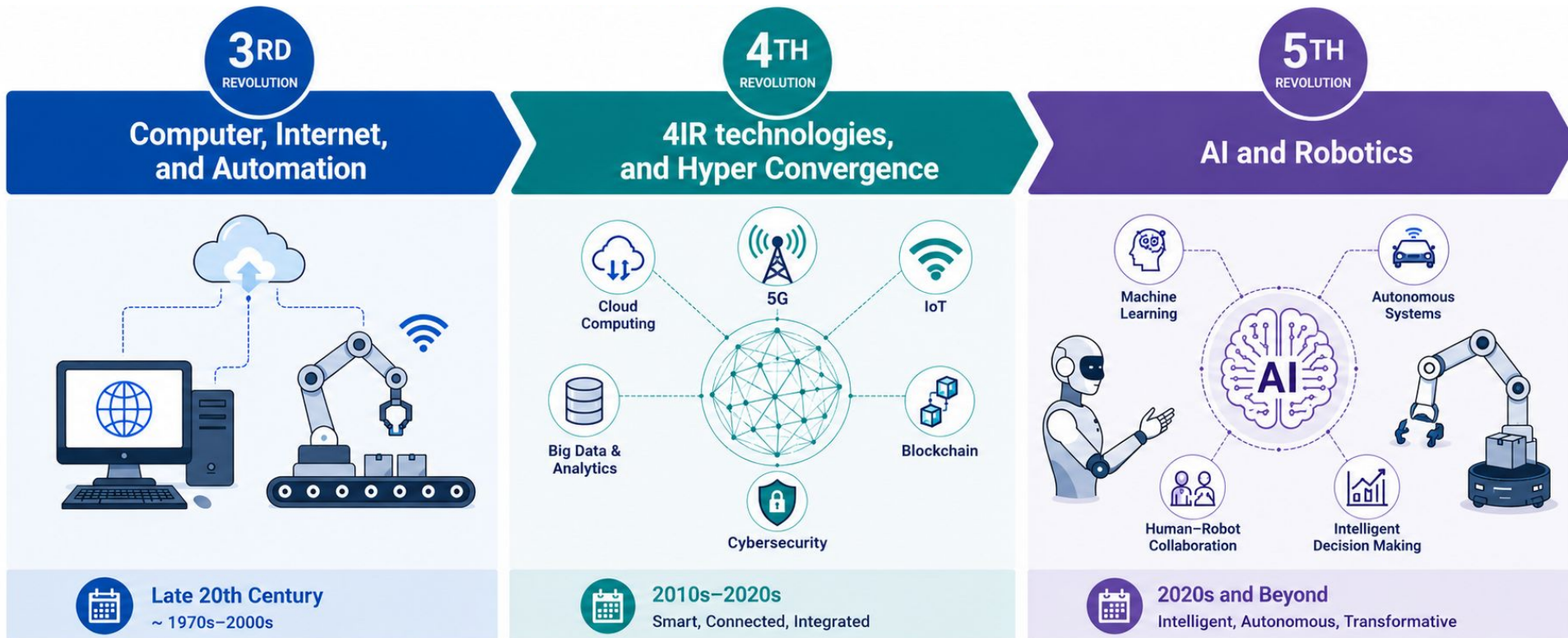
What Is A Smart City?

A Smart City is an urban model that uses Information and Communication Technologies (ICT), such as networks, sensors, and AI, to solve urban challenges, improve quality of life, and enhance economic competitiveness.



The Evolution of Smart Cities

Smart cities have evolved alongside the Industrial Revolutions.



Understanding Robotics

Current robot classification remain largely engineering-driven and industry-oriented.

Form

(FONG et al, 2003)

Humanoid



Bio-inspired



Mechanical



Mobility/Role

(World Robotics Report, 2023)

Locomotion

Ground-Based (rolling, walking)

Aerial (Fly, Hover)

Wearables (Exoskeletons)

Others

Task Types

Professional Cleaning

Inspection & Maintenance

Search, Rescue & Security

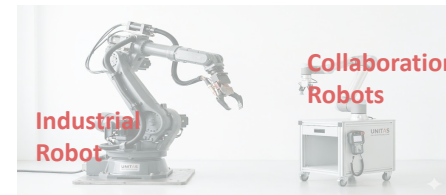
Hospitality

Others..

Industry

(ISO 8373, 2021)

Industrial Robots



Service Robots

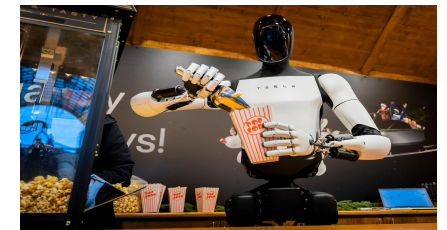


Function

Special Purpose Robots



General Purpose Robots



Robotics in Built Environments

A Governance-Oriented Robot Classification Is Needed.

Industrial Robotics



- Controlled Environments
- Industrial Safety Regulations & Occupational Health Standards
- Interact with Manufacturing Execution System

Building Robotics



- Semi Controlled Environments
- Building codes & standards
- Interact with Building Management Systems (BMS)

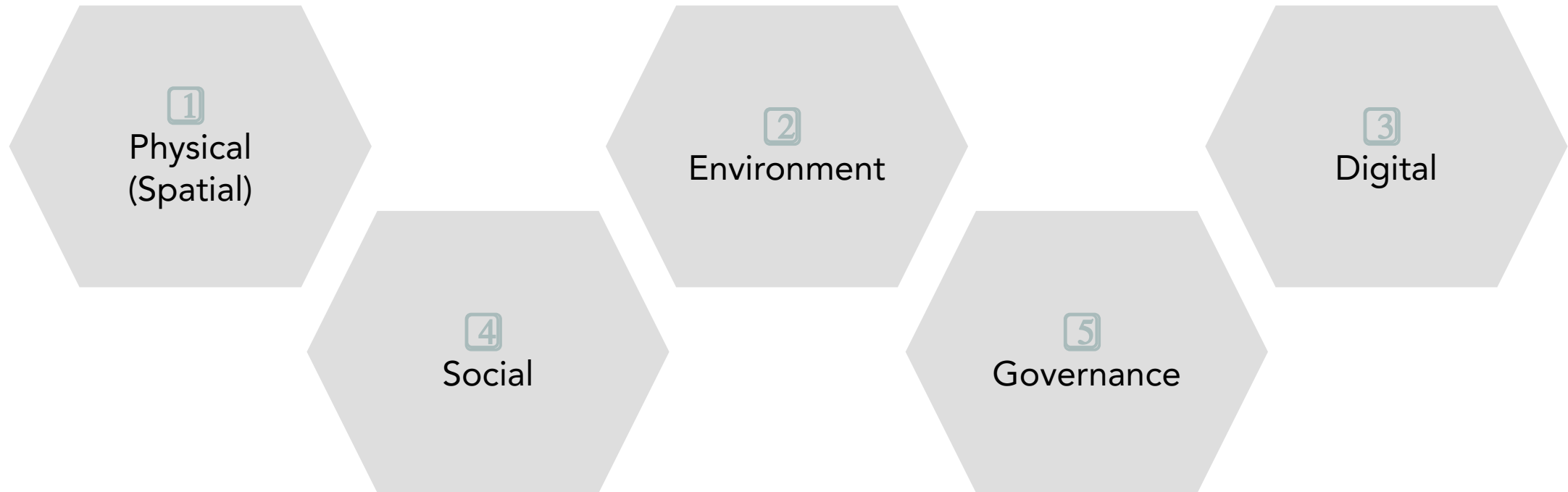
Urban Robotics



- Open & Dynamic Environments
- Public Safety Regulations
- Interact with Smart City Platforms

5 Dimensions for RF City Design

Essential Considerations for Robot Friendly Environments



① Physical Environment

Spatial conditions required to support robot operation in urban environments



The robot from Florida-based Team IHMC Robotics takes a tumble as it tries to walk over rubble. This team came in second place of DARPA robotics challenge (2012-2015) and won a \$1 million prize.

Spatial Barriers

- Narrow and inconsistent sidewalk widths
- High curbs, level gaps, and steep slopes
- Aging infrastructure (cracks, obstacles)
- Disconnected building–street interfaces

Design Considerations for Robots

- Robot-Ready Streets (sufficient sidewalk width, reduced curbs, adequate surface friction)
- Urban-Building Interface Standardization
- Micro-Mobility Integration (mixed flow design, separation/shared rules)
- Urban Logistics Nodes (docking and charging stations)

② Environmental Impacts

Environmental conditions affect robot performance



DEEP Robotics Launches the World's First All-Weather Humanoid Robot (October 17, 2025)

Environmental Considerations

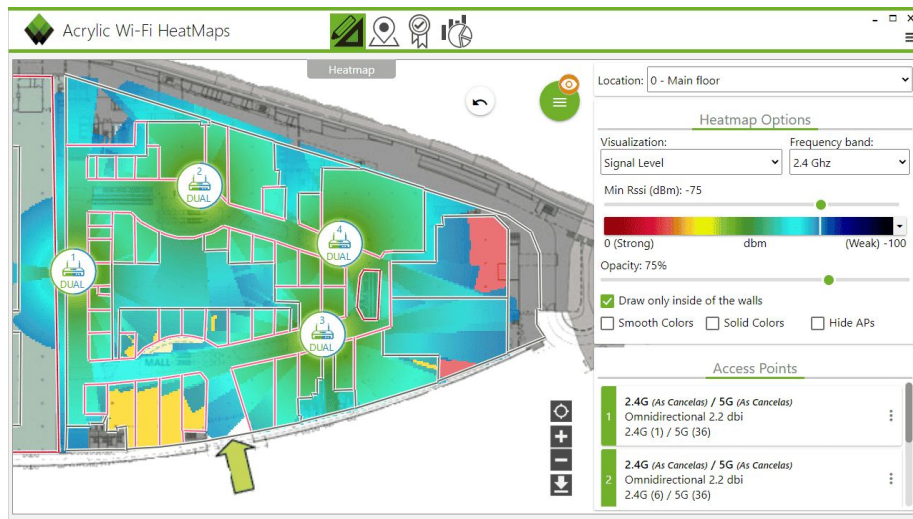
- Weather exposure (rain, snow, heat islands)
- Limited environmental data availability
- Insufficient consideration of ecological impacts

Design Considerations for Robots

- Climate Adaptive Design (sidewalk safe from rain and snow, heat mitigation)
- Blue-Green Infrastructure Integration (Strategies for and deployment of environmental monitoring robots)
- Environmental Sensing Grid (Integration of robots with urban environmental data such as air quality, noise etc.)
- Ecological Impact Control (Guidelines to minimize disturbance to wildlife and ecosystems)

③ Digital Infrastructure

Digital infrastructures necessary to enable reliable robot operation in cities



WiFi Heatmap Assessment Example
(Image: <https://www.tarlogic.com>)

Barriers

- Uneven connectivity (dead zones)
- Fragmented and non-standardized urban data
- Lack of real-time map updates
- Limited interoperability among robot systems

Design for Robots

- Connectivity Infrastructure (Minimize dead zone)
- Urban Digital Twin Integration (real-time map updates)
- Open Data & API Ecosystem (interoperability across robot platforms)
- Fleet Management Layer (city-level robot operation platforms)

④ Social (Human-Robot] Interaction

Urban design facilitating safe and acceptable human–robot coexistence



The German researchers have developed a humanoid robot and a robotic dog designed to support children when crossing the road.

(24 March 2026) <https://www.belganewsagency.eu/>

Barriers

- Pedestrian-only rules dominate urban systems
- Low public acceptance and trust in robots
- Safety concerns and collision risks
- Limited consideration for vulnerable groups

Design for Robots

- Coexistence Protocols (speed limits)
- Public Acceptance Design (robot appearance and behaviors)
- Inclusive access (assistive services such as guidance, delivery support)
- Citizen Engagement (feedback platforms, trainings)

⑤ Governance

Governance frameworks to regulate and coordinate robot integration in urban systems



Smoke billows as a Waymo car burns in the street during protests in Los Angeles on June 8, 2025.

Barriers

- Regulatory gaps
- Unclear allocation of responsibility among stakeholders
- Lack of clear standards for data ownership and usages
- Insufficient public-private collaboration frameworks

Design for Robots

- Regulatory Sandbox & Zoning
- Certification System
- Liability Frameworks
- Data governance (ownership and usage rights of robot-collected data)
- PPP & Operational Model (partnership structure & sustainable revenue models)

RF City Design Principles

Minimize uncertainty and unstructured conditions

Dimension	Key Question	Design Focus	Core Strategy
From Unstructured to Structured Environments	How can urban uncertainty be managed?	Environmental structuring	Transform unstructured spaces into semi-structured systems through design and data
From Objects to Systems	How can robots be integrated as part of urban systems?	System integration	Connect robots with urban infrastructure, data platforms, and operational systems
From Operation to Coexistence	Can robots operate and safely coexist with humans?	Spatial clarity & interaction rules	Reduce ambiguity in movement, define priority rules, ensure predictable interactions
From Innovation to Governance	How can robot integration be governed responsibly?	Institutional frameworks	Establish standards, accountability, and coordinated governance mechanisms

Urban Robotics in Future TOD

A key urban hub for robot mobility, services, and operations.

Robot-Ready Spatial Design

- Robot-Ready Sideway Networks
- Flexible Curb Space Design
- Robot–Human Shared Spaces

Integrated Mobility & Logistics Design

- Robot-Integrated Transit Hubs
- Urban Logistics Hubs
- Seamless Indoor–Outdoor Connectivity

Robot-Supporting Digital Infrastructure

- Robot Charging & Maintenance Infrastructure
- Robot Navigation & Digital Infrastructure
- Dedicated Robot Waiting & Staging Spaces

Robot-Human Interaction Consideration

- Social Acceptance & Inclusive Design
- Human-Robot Behavioral Protocols
- Safety, Ethics & Governance

Conclusion

- Rethinking Cities for Urban Robots
- Five Dimensions of Robot-Friendly Cities
- TOD as an Urban Robotics Hub